INCH-POUND

MIL-M-38510/101G
AMENDMENT 3
17 NOVEMBER 1998
SUPERSEDING
MIL-M-38510/101G
AMENDMENT 2
02 DECEMBER 1994

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, OPERATIONAL AMPLIFIERS,
MONOLITHIC SILICON

Inactive for new design after 13 July 1995

This amendment forms a part of MIL-M-38510/101G, dated 8 May 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1/, line 2, delete “dice” and substitute “die”.

PAGE 3

3.1, delete and substitute the following;
“3.1 Item requirements. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. The individual item requirements for devices marked with the “Q” certification mark shall be in accordance with MIL-I-38535 and as specified herein or as modified in the device manufacturer’s Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.”

PAGE 4

Table I, add new footnote “1/” after the title which will state the following;
“1/ For devices marked with the “Q” certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer’s QM plan.”

Table I, input offset voltage temperature sensitivity, symbol column, delete “$\Delta V_{IO}$” and substitute “$\frac{\Delta V_{IO}}{\Delta T}$”. Also, footnote “1/” will be renumbered as footnote “2/”.

Table I, input offset current temperature sensitivity, symbol column, delete “$\Delta I_{IO}$” and substitute “$\frac{\Delta I_{IO}}{\Delta T}$”. Also, footnote “1/” will be renumbered as footnote “2/”.

Table I, input bias current test, footnote “1/” will be renumbered as footnote “2/”. Device types 01, 02, 03, 05, 07, and 08, delete “+1 nA min” and substitute “-0.1 nA min”.


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TABLE I, add new footnote "1/" after the title which will state the following;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

TABLE I, Adjustment for input offset voltage (positive) test, footnote "2/" will be renumbered as footnote "3/".

TABLE I, Adjustment for input offset voltage (negative) test, footnote "2/" will be renumbered as footnote "3/".

TABLE I, Output short-circuit current (for positive output) test, footnote "3/" will be renumbered as footnote "4/".

TABLE I, Output short-circuit current (for negative output) test, footnote "3/" will be renumbered as footnote "4/". Also, delete in its entirety and substitute the following:

| Characteristics | Symbol | Conditions (Figure 4, unless otherwise specified | | Limits | | | | | | | | Units |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 01,02,08 | | 03,05 | | 04,06 | | 07 | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| Output short-circuit (for negative output) | $I_{OS-}$ | $\pm V_{CC}$ = 15 V, $T_A$ = -55°C to +125°C t ≤ 25 ms 4/ | | | +60 | | +60 | | +15 | | | mA |
| | | | $T_A$ = +25°C to+125°C | | | | | | | | +65 | mA |
| | | | $T_A$ = -55°C | | | | | | | | +80 | mA |

TABLE I, Supply current test, footnote "4/" will be renumbered as footnote "5/".

TABLE I, Both open loop voltage gain (single ended) tests, footnote "5/" will be renumbered as footnote "6/". Also, footnote "6/" will be renumbered as footnote "7/".

TABLE I, Transient response rise time tests, footnote "7/" will be renumbered as footnote "8/".

TABLE I, Transient response overshoot tests, footnote "7/" will be renumbered as footnote "8/".

TABLE I, Both slew rate tests, footnote "8/" will be renumbered as footnote "9/". Also, footnote "9/" will be renumbered as footnote "10/".

* TABLE I, Slew rate test, SR(+), for device type 07, delete "+50" and substitute "+40". Also, delete "+40" and substitute "+30".

* TABLE I, Slew rate test, SR(-), for device type 07, delete "+50" and substitute "+40". Also, delete "+40" and substitute "+30".



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PAGE 6

TABLE I, Settling time test, footnote "10/" will be renumbered as footnote "11/". Also, limits column for device type 07, delete in its entirety and substitute the following:

| Characteristics | Symbol | Conditions (Figure 4, unless otherwise specified | Limits | | | | | | | | Units |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 01,02,08 | | 03,05 | | 04,06 | | 07 | | |
| | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| Settling time 11/ | $t_{S+}$ | See figure 5, $T_A$ = +25°C | | | | | | | | 800 | ns |
| | | $T_A$ = -55°C to +125°C | | | | | | | | 1200 | ns |
| | $t_{S-}$ | See figure 5, $T_A$ = +25°C | | | | | | | | 800 | ns |
| | | $T_A$ = -55°C to +125°C | | | | | | | | 1200 | ns |

Add a new footnote "1/" which will state;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

Footnote "1/" will be renumbered as footnote "2/".

Footnote "2/" will be renumbered as footnote "3/".

Footnote "3/" will be renumbered as footnote "4/".

Footnote "4/" will be renumbered as footnote "5/".

Footnote "5/" will be renumbered as footnote "6/".

Footnote "6/" will be renumbered as footnote "7/".

Footnote "7/" will be renumbered as footnote "8/".

Footnote "8/" will be renumbered as footnote "9/".

Footnote "9/" will be renumbered as footnote "10/".

Footnote "10/" will be renumbered as footnote "11/".

PAGE 7

4.1, delete and substitute the following;
"4.1 Sampling and inspection. Sampling inspection procedures shall be in accordance with MIL-M-38510 and methods 5004, 5005, and 5007, as applicable, of MIL-STD-883, except as modified herein. For devices marked with the "Q" certification mark, sampling and inspection procedures shall be in accordance with MIL-I-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein."



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PAGE 8

4.4.2b, delete the last sentence and substitute the following; “If alternate burn-in conditions are used, the circuit shown in either Figure 7 or Figure 8 shall be used.”

4.2.d, line 3, delete “either”.

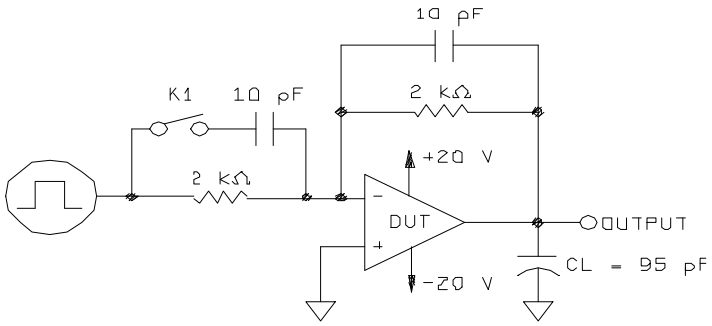
4.4.3.c, lines 3 and 4, delete and substitute the following:
* “The sample size series for subgroup 12 shall be 5, and 13 shall be 7 for class B devices (see MIL-PRF-38535, Appendix D).”

PAGE 18

FIGURE 4 at the center upper half of the circuit at $I_{OS}$ meter, delete “$S_7$” and substitute “$S_6$”. Also, at the center bottom of the circuit $S_{4A}$ position 7, delete the “200 Ω” resistor and substitute “200 kΩ”.

6/, delete and substitute the following; “6/ Device type 01, all case types, and device type 02, case A, B, C, and D only.”

FIGURE 5, Add the following diagram as an alternate test circuit for device type 07 only.




NOTES:
1. K1 is closed for small signal tests (Tr and P.O.) and is open for large signal tests (±slew rate, ±Ts).
2. Input signal is a -50 mV to 0 mV pulse train for small signal tests and -5 V to +5 V pulse train for large signal tests.
3. Tr of the input signal is < 10 ns for the small signal tests.

PAGE 24

FIGURE 5, add note: “9. $C_L$ capacitance specified includes stray, jig, and probe capacitance.”

PAGE 27

FIGURE 8. Note 2, delete and substitute the following; “For device types 01 and 04, $V_{IN}$ = 3.0 V dc. For device type 03, $V_{IN}$ = +3 V or -3 V.”



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PAGE 30

TABLE III, test numbers 9-16, device types 01, 02, 03, 05, 07, and 08, delete “+1 nA min” and substitute “-0.1 nA min”.

PAGE 31

TABLE III, test numbers 35-42, device types 01, 02, 03, 05, 07, and 08, delete “+1 nA min” and substitute “-0.1 nA min”.

PAGE 32

TABLE III, test numbers 61-68, device types 01, 02, 03, 05, 07, and 08, delete “+1 nA min” and substitute “-0.1 nA min”.

PAGE 33

Table III, test numbers 81 and 83, conditions column, delete “-15 V” and substitute “+15 V”.

Table III, test numbers 82 and 84, conditions column, delete “+15 V” and substitute “-15 V”.

Table III, test number 86, conditions column, delete “2 kΩ” and substitute “10 kΩ”.

PAGE 34

Table III, test numbers 91, 93, 101, and 103, conditions column, delete “-15 V” and substitute “+15 V”.

Table III, test numbers 92, 94, 102, and 104, conditions column, delete “+15 V” and substitute “-15 V”.

Table III, test numbers 96 and 106, conditions column, delete “2 kΩ” and substitute “10 kΩ”.

* Table III, test numbers 109 and 110, device type 07, delete “50” and substitute “40”.

Table III, test number 110, conditions column, delete “$V_{IN}$ = -5 V to +5 V” and substitute “$V_{IN}$ = +5 V to -5 V”.

PAGE 35

* Table III, test numbers 116 and 117, device type 07, delete “40” and substitute “30”.

* Table III, test numbers 120 and 121, device type 07, delete “50” and substitute “40”.

TABLE III, add a sentence to footnote "1/" which will state the following;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

3/, delete and substitute the following; “$V_{IN}$(adj) is not performed on device type 02, case I only, or on device types 04, 06, and 08, all case types.”



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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Air Force - 19, 85, 99
Navy - AS, CG, MC, OS, SH

Civil Agency Coordinating Activity:
DOT-FAA(RD-650)

Preparing activity:
DLA - CC

(Project 5962-1827)



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